Slides, posters, and figures

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Talks, posters, and figures carry the same research as a paper, but the reader sees them for less time and cannot reread them. This chapter gives practical rules for each.

Guidelines for slide decks

When preparing slides for a lab meeting, conference, qualifying exam, or thesis defense, follow these guidelines:

- **Always include slide numbers**: Slide numbers make feedback specific. During questions or a practice run, listeners need to refer back to a slide quickly (e.g., "On slide 14, how was that variable defined?") instead of describing it or guessing where a topic occurred.
- **Complete, record, and review practice runs**: Before any significant presentation, rehearse the complete talk out loud, record the practice run (with screen recording software, a webcam, or your phone), and watch the recording. The recording shows your pacing, timing, filler words, and transitions, and whether slide text and figures stay legible on screen.
- **One main point per slide**: Limit the cognitive load of each slide. If a slide presents several findings or a complex argument, split it across consecutive slides so the audience can follow each point.
- **Prefer visual evidence**: Audiences cannot read dense paragraphs while listening to a speaker. Use clean charts, diagrams, and annotated figures instead of text-heavy bullet lists.
- **Ensure high contrast and readability**: Use large fonts and high-contrast colors. Check that axis labels, legends, and annotations can be read from the back of a large room or on a low-resolution projector.

Presentation and poster resources

- How to tell a compelling story in scientific presentations (Kirchoff 2021)
- The science of storytelling: the David Attenborough style of scientific presentation (Ratcliff 2023)
- How to create a better research poster in less time - Mike Morrison's video introducing the #betterposter template
- Towards an "#evenbetterposter" - the MIT Biological Engineering Communication Lab's improvements to that template

Scientific figures

A clear figure lets a reader check a result without help from the text. Before you share a figure:

- Label the x-axis and y-axis clearly.
- Use panel labels when a figure has several subplots.
- Include enough tick marks and tick labels.
- Print tick labels as plain or comma-separated numbers, such as 0.001 or 10,000, not in scientific notation. A logarithmic axis is fine. Only the label format matters, so write 1, 10, 100 and 1,000, not 1e+00 through 1e+03. When the numbers are huge or tiny, rescale the units and say so in the axis title, such as "dollars (millions)".
- Use font sizes that are easy to read without dominating the plot.
- Use colorblind-friendly palettes (see ColorBrewer or iwanthue).
- Define colors, shapes, and line types in a legend.
- Minimize abbreviations in labels and legends.
- Use transparency to show overlapping data.
- Show measures of precision, such as 95% confidence intervals.

For more:

- Ten Simple Rules for Better Figures (Rougier et al. 2014)
- Tufte CSS - style HTML articles like Edward Tufte's books and handouts, with sidenotes, typography, and tight integration of graphics with text

Three-dimensional graphs

A function of two variables, such as a loss or a residual sum of squares as a function of a regression's intercept and slope, is a surface. A contour plot shows where the function takes given values and is easy to read values from. A three-dimensional surface plot shows the same function as a landscape, so valleys, ridges and minima are visible at once. Show both when the shape of the surface matters.

- Use a 3D plot for a surface, with the function's value on the vertical axis, and pair it with a contour plot of the same function.
- Do not use 3D for data that are not a surface, such as bars or pie charts: perspective makes lengths and areas harder to compare.
- Choose the viewing angle so that the feature you discuss, such as the minimum, is not hidden, and label all three axes.
- An interactive plot that readers can rotate, such as one made with plotly, helps on the web, but PDF and Word output show no interactive plot, so also provide a static view at a fixed angle (see Make figures accessible).
- Give a 3D plot alt text that states what the surface looks like, such as "a bowl with its lowest point at …".

Source

The sections on slide decks, presentation resources and scientific figures are adapted from the Professional Development and Resources chapters of the UCD-SERG lab manual. Several of the resource links began in the Benjamin-Chung Lab Manual by Jade Benjamin-Chung. The section on three-dimensional graphs is original to this guide.

Kirchoff, Bruce. 2021. "How to Tell a Compelling Story in Scientific Presentations." *Nature* 600 (7890): S88–89. https://doi.org/10.1038/d41586-021-03603-2.

Ratcliff, William C. 2023. "The Science of Storytelling: The David Attenborough Style of Scientific Presentation." *Trends in Molecular Medicine* 29 (9): 675–77. https://doi.org/10.1016/j.molmed.2023.05.002.

Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. 2014. "Ten Simple Rules for Better Figures." *PLOS Computational Biology* 10 (9): e1003833. https://doi.org/10.1371/journal.pcbi.1003833.